



Audyogik Shikshan Mandals Nextgen Technical Campus

Survey No. 401/1, Adjacent to Parshwa Prandyalay Jain Temple Talegaon Dabhade,
Maval, Pune -410 506

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Affiliated to SPPU and Approved by AICTE

Unit Wise Question Bank

Department: MCA

Semester: I/II

Academic Year: 2025 - 2026

Class: FYMCA

Div: A, B, C

Date: 4/05/2026

Course: Optimization Techniques

Unit No. - name	Q. No.	Topic name	Questions	Year	Marks																													
1- Linear Programming Problems	Q.1	Two Phase Simplex Method	Solve with two phase simplex method Maximize $Z = 3X1 + 4X2$ Subject to, $2X1+3X2 \leq 200$ $5X1+4X2 \geq 100$ $- 8X1 - 4X2 \leq -80$ Such that $X1, X2 \geq 0$	2024	8																													
	Q.2	Transportation Problem	A company has three plants W, X and Y and three warehouses A, B, C. The supplier is transported from the plants to the warehouses which are located at varying distances from the plants. Due to this, the transportation costs from plants to warehouses vary from Rs. 8 to Rs. 24 per unit. <table border="1"><tr><td>Plant</td><td colspan="3">Warehouse</td><td rowspan="2">Supply</td></tr><tr><td></td><td>A</td><td>B</td><td>C</td></tr><tr><td>W</td><td>12</td><td>8</td><td>18</td><td>400</td></tr><tr><td>X</td><td>20</td><td>10</td><td>16</td><td>350</td></tr><tr><td>Y</td><td>24</td><td>14</td><td>12</td><td>300</td></tr><tr><td>Demand</td><td>500</td><td>200</td><td>300</td><td></td></tr></table>	Plant	Warehouse			Supply		A	B	C	W	12	8	18	400	X	20	10	16	350	Y	24	14	12	300	Demand	500	200	300		2024	8
	Plant	Warehouse			Supply																													
		A	B	C																														
	W	12	8	18	400																													
X	20	10	16	350																														
Y	24	14	12	300																														
Demand	500	200	300																															
Q.3	Terminology of LPP	Explain the following (any one) i) Infeasible Solution ii) Slack and Surplus Variable	2024	2																														
Q.4.	Maximization Simplex Method	Solve the following using simplex method. Maximize $Z = 3x1 + 2x2 + 5x3$ Subject to: $x1+ x2 + x3 \leq 9,$ $2x1 + 3x2 + 5x3 \leq 20,$ $2x1 - x2 - x3 \leq 8,$ & $x1, x2, x3 \geq 0$	2019	10																														
Q.7	Maximization Simplex Method	Solve following LPP Maximize $Z = 50x_1 + 100x_2$ Subject to: $x_1 + x_2 \leq 150$, $2x_1 + 3x_2 \leq 360$, $x_1, x_2 \geq 0$	2020-I	6																														



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	Q.8	Maximization Simplex Method	Solve following LPP Maximize $Z = 7x_1 + 6x_2$ Subject to: $x_1 + x_2 \leq 4$ $2x_1 + x_2 \leq 6$ $x_1, x_2 \geq 0$	2020-I	6																							
	Q.9	Maximization Simplex Method	Use the simplex method to solve the following LP problem. Maximize $Z = 3x_1 + 5x_2 + 4x_3$ Subject to: $2x_1 + 3x_2 \leq 8$ $2x_2 + 5x_3 \leq 10$ $3x_1 + 2x_2 + 4x_3 \leq 15$ and $x_1, x_2, x_3 \geq 0$	2020-II	10																							
	Q.10	Two Phase Simplex Method	Solve the following L.P.P. by two-phase method Maximize $Z = 4x + 3y$ Subject to: $2x + 3y \leq 6$ $3x + y \geq 3$ and $x, y \geq 0$	2020-II	10																							
	Q.11	Transportation Problem	Solve the following transportation problem to find optimum solution and cost associated with it: <table border="1"><thead><tr><th>Sources</th><th>Destination 1</th><th>Destination 2</th><th>Destination 3</th><th>Destination 4</th><th>Supply</th></tr></thead><tbody><tr><td>1</td><td>11</td><td>13</td><td>17</td><td>14</td><td>250</td></tr><tr><td>2</td><td>16</td><td>18</td><td>14</td><td>10</td><td>300</td></tr><tr><td>3</td><td>24</td><td>12</td><td>13</td><td>10</td><td>400</td></tr></tbody></table> Demand: 200, 225, 275, 250	Sources	Destination 1	Destination 2	Destination 3	Destination 4	Supply	1	11	13	17	14	250	2	16	18	14	10	300	3	24	12	13	10	400	2019
Sources	Destination 1	Destination 2	Destination 3	Destination 4	Supply																							
1	11	13	17	14	250																							
2	16	18	14	10	300																							
3	24	12	13	10	400																							
Q.12	Maximization Simplex Method	Solve using simplex method: Maximize $Z = 3x_1 + 2x_2 + 5x_3$ Subject to: $x_1 + x_2 + x_3 \leq 9$ $2x_1 + 3x_2 + 5x_3 \leq 20$ $2x_1 - x_2 - x_3 \leq 8$ $x_1, x_2, x_3 \geq 0$	2019	10																								



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2-Sequential model, Queuing theory and related problems

Q.13

Job Sequencing Problem- Processing n jobs through 3 machines

5 Jobs are to be processed through 3 machines. Processing time (in hrs) are as follows:

Determine:

i) Sequence of jobs.

ii) Elapsed time for machine A, B, &C

Jobs	1	2	3	4	5
Machine-A	10	11	8	7	6
Machine-B	6	4	5	3	2
Machine-C	9	5	4	6	8

2020-I

6

Q.14

Job Sequencing Problem- Processing n jobs through 2 machines

Seven jobs are to be processed through 2 machines A & B processing time (in hrs) are as follows.

Jobs	1	2	3	4	5	6	7
Machine-A	10	9	7	15	18	20	14
Machine-B	12	8	7	12	10	6	13

Determine:

i) Sequence of jobs.

ii) Idle time for machine A & B.

2020-I

6

Q.15

Processing n jobs through 3 machines

Jobs	1	2	3	4	5	6
Machine A	8	3	7	2	5	2
Machine B	3	4	5	2	1	6
Machine C	8	7	2	10	9	9

ix jobs go first on **Machine A**, then **Machine B**, and then **Machine C**. The processing times (hours) are given below:

Task:

- Find the sequence of jobs that minimizes total elapsed time to complete all jobs.
- Also find idle time of all 3 machines.

2019

10

Q.16

Processing n jobs through 2 machines

There are 7 jobs each of which must go through three machines M1, M2 and M3. The corresponding processing time on the machines in hours are as follows

Job	J1	J2	J3	J4	J5	J6	J7
M1	3	8	7	4	9	8	7
M2	4	3	2	5	1	4	3
M3	6	7	5	11	5	6	12

i) What is the optimal job sequence that will minimize the total elapsed time?

ii) What is the idle time for each machine (M1, M2 and M3)?

2024

5

	Q.17	Processing n jobs through 2 machines	There are seven jobs, each of which has to go through machines A and B in the order BA. Processing time in hours as follows: Determine an optimum sequence of these jobs that will minimize the total elapsed time (T). Also find T and calculate idle time for machine A and B. <table><tr><td>Job</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>Machine A</td><td>3</td><td>12</td><td>15</td><td>6</td><td>10</td><td>11</td><td>9</td></tr><tr><td>Machine B</td><td>8</td><td>10</td><td>10</td><td>6</td><td>12</td><td>1</td><td>3</td></tr></table>	Job	1	2	3	4	5	6	7	Machine A	3	12	15	6	10	11	9	Machine B	8	10	10	6	12	1	3	2024	5				
Job	1	2	3	4	5	6	7																										
Machine A	3	12	15	6	10	11	9																										
Machine B	8	10	10	6	12	1	3																										
	Q.18	Queuing Theory	In a bank with a single cashier, customers arrive on average every 20 minutes and the cashier takes 10 minutes to attend to each customer. i) What is the average number of customers waiting for the cashier? ii) What is the average time spent by each customer in the bank?																														
	Q.19	Processing n jobs through 3 machines	A machine operator has to perform three operations: plane turning, step turning, and taper turning on different jobs. The time required (minutes) for each operation per job is given below. Find the optimal sequence that minimizes total time. <table><tr><td>Job</td><td>Plane turning</td><td>Step turning</td><td>Taper turning</td></tr><tr><td>1</td><td>3</td><td>8</td><td>13</td></tr><tr><td>2</td><td>12</td><td>6</td><td>4</td></tr><tr><td>3</td><td>5</td><td>4</td><td>9</td></tr><tr><td>4</td><td>8</td><td>6</td><td>12</td></tr><tr><td>5</td><td>4</td><td>9</td><td>13</td></tr><tr><td>6</td><td>11</td><td>3</td><td>7</td></tr></table>	Job	Plane turning	Step turning	Taper turning	1	3	8	13	2	12	6	4	3	5	4	9	4	8	6	12	5	4	9	13	6	11	3	7	2020-II	7
Job	Plane turning	Step turning	Taper turning																														
1	3	8	13																														
2	12	6	4																														
3	5	4	9																														
4	8	6	12																														
5	4	9	13																														
6	11	3	7																														
	Q.20	Queuing Theory	Customers arrive at a one-window drive-through pharmacy according to a Poisson distribution with mean 10 per hour. Service time per customer is exponential with mean 5 minutes. Find: a) Probability that a customer has to wait b) Number of customers in the system c) Average waiting time of customer before getting service	2019	10																												
	Q.21	Queuing Theory	The person works in the internet café works for 8 hours a day. The rate of arrival of customers at an internet café follows Poisson distribution with average of fifteen minutes between one customer and the next. The duration of service is assumed to follow exponential distribution with mean of twelve minutes. i) What is the probability that person arriving at the café will have to wait? ii) Find the total time for which the person will be idle.	2024	5																												
	Q.22	Processing n jobs through 2 machines	A readymade garments manufacturer has to process 7 items through two stages of production: cutting and sewing. The time taken for each item is given below: <table><tr><td>Item</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>Cutting (time units)</td><td>5</td><td>7</td><td>3</td><td>4</td><td>6</td><td>7</td><td>12</td></tr><tr><td>Sewing (time units)</td><td>2</td><td>6</td><td>7</td><td>5</td><td>9</td><td>5</td><td>8</td></tr></table> Find an order in which these items are to be processed through these stages so as to	Item	1	2	3	4	5	6	7	Cutting (time units)	5	7	3	4	6	7	12	Sewing (time units)	2	6	7	5	9	5	8	2020-II	7				
Item	1	2	3	4	5	6	7																										
Cutting (time units)	5	7	3	4	6	7	12																										
Sewing (time units)	2	6	7	5	9	5	8																										

			minimize the total processing time.																					
3-Markovs Chain, Simulation Techniques & Inventory Theory	Q.23	Markov Chains-Transition Problem	Market share of brands A, B, & C are 50%, 30% & 20%. Customers shift their brands in the matrix as below: <table><tr><th rowspan="2">From</th><th colspan="3">To</th></tr><tr><th>A</th><th>B</th><th>C</th></tr><tr><th>A</th><td>50%</td><td>30%</td><td>20%</td></tr><tr><th>B</th><td>20%</td><td>70%</td><td>10%</td></tr><tr><th>C</th><td>20%</td><td>20%</td><td>60%</td></tr></table> Find: i) Transition matrix for brands. ii) Find share at the end of period 1&2	From	To			A	B	C	A	50%	30%	20%	B	20%	70%	10%	C	20%	20%	60%	2020-I	4
	From	To																						
		A	B	C																				
	A	50%	30%	20%																				
	B	20%	70%	10%																				
C	20%	20%	60%																					
Q.24	Markov Chain-Multi Period Transition Problem	A market survey is made on 3 brands of breakfast foods x, y, z. Every time a customer purchases a new package, the following estimates for shift (in percent) are obtained: <table><tr><th></th><th colspan="3">Next</th></tr><tr><th>Present</th><th>X</th><th>Y</th><th>Z</th></tr><tr><th>X</th><td>70</td><td>20</td><td>10</td></tr><tr><th>Y</th><td>30</td><td>50</td><td>20</td></tr><tr><th>Z</th><td>30</td><td>30</td><td>40</td></tr></table> It is estimated that 30% of people buy brand x, 20% buy brand y, and 50% buy brand z. What will be the distribution of the customers for the first and second period?		Next			Present	X	Y	Z	X	70	20	10	Y	30	50	20	Z	30	30	40	2020-I	4
	Next																							
Present	X	Y	Z																					
X	70	20	10																					
Y	30	50	20																					
Z	30	30	40																					
Q.25	Monte Carlo Simulation Model	Distribution of demand for books has been found following information. <table><tr><td>Demand</td><td>15</td><td>16</td><td>17</td><td>18</td><td>19</td><td>20</td><td>21</td><td>22</td></tr><tr><td>No. of Copies</td><td>0.05</td><td>0.08</td><td>0.20</td><td>0.45</td><td>0.10</td><td>0.07</td><td>0.03</td><td>0.02</td></tr></table> Generate demand for 20 times period using following random numbers, also calculate average no. of copies demand. 14, 2, 93, 99, 18, 71, 37, 30, 12, 10, 88, 13, 00, 57, 69, 32, 18, 8, 92, 73	Demand	15	16	17	18	19	20	21	22	No. of Copies	0.05	0.08	0.20	0.45	0.10	0.07	0.03	0.02	2020-I	4		
Demand	15	16	17	18	19	20	21	22																
No. of Copies	0.05	0.08	0.20	0.45	0.10	0.07	0.03	0.02																
Q.26	Monte Carlo Simulation Model	A confectioner sells items past data of demand per week is given below. <table><tr><td>Demand in (kg)</td><td>0</td><td>5</td><td>10</td><td>15</td><td>20</td><td>25</td></tr><tr><td>Probability</td><td>0.04</td><td>0.22</td><td>0.16</td><td>0.42</td><td>0.10</td><td>0.06</td></tr></table> Generate demand for next 10 weeks & average demand per week using random numbers - 78, 99, 43, 62, 44, 2, 67, 32, 54, 75	Demand in (kg)	0	5	10	15	20	25	Probability	0.04	0.22	0.16	0.42	0.10	0.06	2020-I	4						
Demand in (kg)	0	5	10	15	20	25																		
Probability	0.04	0.22	0.16	0.42	0.10	0.06																		
Q.27	Multi period transition	Two manufacturers A and B are competing in a restricted market. A’s customers show 80% loyalty (continue with A).	2020-II	7																				

		probabilities	- Former customers of B switch back to A 60% of the time. i) Construct and interpret the state transition matrix in terms of: - Retention and loss - Retention and gain ii) Calculate the probability of a customer purchasing A’s product at the end of the second period.																		
	Q.28	Multi period transition probabilities	Three companies introduced new razor blades in the market at the same time, each initially holding an equal market share. After the first year, some changes occurred, which are shown by the following transition matrix: <table><tr><td></td><td>A</td><td>B</td><td>C</td></tr><tr><td>A</td><td>0.9</td><td>0.03</td><td>0.07</td></tr><tr><td>B</td><td>0.2</td><td>0.6</td><td>0.2</td></tr><tr><td>C</td><td>0.1</td><td>0.2</td><td>0.7</td></tr></table> Assume no changes in customer buying habits occur. What will be the market share of each company at the end of the first and second year?		A	B	C	A	0.9	0.03	0.07	B	0.2	0.6	0.2	C	0.1	0.2	0.7	2024	5
	A	B	C																		
A	0.9	0.03	0.07																		
B	0.2	0.6	0.2																		
C	0.1	0.2	0.7																		
	Q.29	Multi period transition probabilities	Two companies, A, B compete against each other. The transition matrix T for people switching each month among them is given by the following transition matrix: <table><tr><td colspan="3">NEXT</td></tr><tr><td>INITIAL</td><td>A</td><td>B</td></tr><tr><td>A</td><td>65%</td><td>35%</td></tr><tr><td>B</td><td>45%</td><td>55%</td></tr></table> Find the long term distribution	NEXT			INITIAL	A	B	A	65%	35%	B	45%	55%	2024	5				
NEXT																					
INITIAL	A	B																			
A	65%	35%																			
B	45%	55%																			
	Q.30	Monte Carlo Simulation method	The daily demand for Coca-Cola bottles, based on past experience, is given as follows: <table><tr><td>Daily Demand</td><td>10</td><td>20</td><td>30</td><td>40</td><td>50</td></tr><tr><td>Probability</td><td>0.15</td><td>0.10</td><td>0.40</td><td>0.20</td><td>0.15</td></tr></table> A manager decides to keep 25 Coca-Cola bottles in stock. Using the following random numbers: 69, 01, 08, 74, 82, 20, 72, 14, 75, 12, 25 estimate the average balance stock.	Daily Demand	10	20	30	40	50	Probability	0.15	0.10	0.40	0.20	0.15	2024	5				
Daily Demand	10	20	30	40	50																
Probability	0.15	0.10	0.40	0.20	0.15																
	Q.31	Monte Carlo Simulation method	A bakery keeps stock of a popular cake. Demand distribution: <table><tr><td>Daily demand</td><td>0</td><td>10</td><td>20</td><td>30</td><td>40</td><td>50</td></tr><tr><td>Probability</td><td>0.01</td><td>0.20</td><td>0.15</td><td>0.50</td><td>0.12</td><td>0.02</td></tr></table> Random numbers: 25, 39, 65, 76, 12, 05, 73, 89, 19, 49. Simulate demand for 10 days and estimate average daily demand.	Daily demand	0	10	20	30	40	50	Probability	0.01	0.20	0.15	0.50	0.12	0.02	2020-II	7		
Daily demand	0	10	20	30	40	50															
Probability	0.01	0.20	0.15	0.50	0.12	0.02															
	Q.32	Monte Carlo Simulation method	The Lajawab Bakery shop keeps stock of a popular brand of cake previous experience indicates the daily demand as given below: <table><tr><td>Daily Demand</td><td>0</td><td>15</td><td>25</td><td>35</td><td>45</td><td>50</td></tr><tr><td>Probability</td><td>0.01</td><td>0.10</td><td>0.30</td><td>0.45</td><td>0.12</td><td>0.02</td></tr></table> Consider the following sequence of random numbers 21, 28, 57, 54, 60, 39, 74, 91, 75, 30. Using this sequence simulate the demand for next 10 days. Find out the	Daily Demand	0	15	25	35	45	50	Probability	0.01	0.10	0.30	0.45	0.12	0.02	2024	5		
Daily Demand	0	15	25	35	45	50															
Probability	0.01	0.10	0.30	0.45	0.12	0.02															



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			stock situation if the owner of the bakery shop decides to make 30 cakes every day.																																	
	Q.33	Multi period transition probabilities	Two manufacturers A and B are competing in a restricted market. - A's customers show 80% loyalty (continue with A). - Former customers of B switch back to A 60% of the time. i) Construct and interpret the state transition matrix in terms of: - Retention and loss - Retention and gain ii) Calculate the probability of a customer purchasing A's product at the end of the second period.	2020-II	7																															
	Q.34		Annual demand of an item is 2,400 units. - Ordering cost: Rs. 350 per order - Inventory carrying cost: 24% of purchase price per year - Price per item: Rs. 9.25 Find: a) Economic order quantity b) Total cost related to EOQ c) Optimum number of orders d) Time between two orders.	2019	10																															
	Q.35	Simulation Techniques	A bakery keeps stock of a popular cake. Demand distribution: <table border="1"><tr><td>Daily demand</td><td>0</td><td>10</td><td>20</td><td>30</td><td>40</td><td>50</td></tr><tr><td>Probability</td><td>0.01</td><td>0.20</td><td>0.15</td><td>0.50</td><td>0.12</td><td>0.02</td></tr></table> Random numbers: 25, 39, 65, 76, 12, 05, 73, 89, 19, 49. Simulate demand for 10 days and estimate average daily demand.	Daily demand	0	10	20	30	40	50	Probability	0.01	0.20	0.15	0.50	0.12	0.02	2020-II	7																	
Daily demand	0	10	20	30	40	50																														
Probability	0.01	0.20	0.15	0.50	0.12	0.02																														
4-PERT & CPM	Q.36	Terminology	Explain the terms. i) Basic solution. ii) Dummy activity	2020-I	4																															
	Q.37	Terminology	Explain the terms: i) Degenerate solution. ii) Float.	2020-I	4																															
	Q.38	Terminology	Explain the following (Any One): a) Difference between PERT and CPM b) Dummy Activity	2020-II	2																															
	Q.39	Network Diagram, Critical Path, Probability	Small project consist 7 activities whose time estimates are as follows: <table border="1"><tr><th>Activity (i - j)</th><th>Optimistic</th><th>Most likely</th><th>Pessimistic</th></tr><tr><td>1-2</td><td>1</td><td>1</td><td>7</td></tr><tr><td>1-3</td><td>1</td><td>4</td><td>7</td></tr><tr><td>1-4</td><td>2</td><td>2</td><td>8</td></tr><tr><td>2-5</td><td>1</td><td>1</td><td>1</td></tr><tr><td>3-5</td><td>2</td><td>5</td><td>14</td></tr><tr><td>4-6</td><td>2</td><td>5</td><td>8</td></tr><tr><td>5-6</td><td>3</td><td>6</td><td>15</td></tr></table> Determine: i) Draw network diagram & identify critical path. ii) Identify critical activities & duration of project. iii) Determine probability of completing project in 13 weeks.	Activity (i - j)	Optimistic	Most likely	Pessimistic	1-2	1	1	7	1-3	1	4	7	1-4	2	2	8	2-5	1	1	1	3-5	2	5	14	4-6	2	5	8	5-6	3	6	15	2020-I
Activity (i - j)	Optimistic	Most likely	Pessimistic																																	
1-2	1	1	7																																	
1-3	1	4	7																																	
1-4	2	2	8																																	
2-5	1	1	1																																	
3-5	2	5	14																																	
4-6	2	5	8																																	
5-6	3	6	15																																	

	Q.40	EMV	An investor is given the following investment alternatives and percentage rates of return. States of Nature (Market Conditions): <table><tr><th>Investment</th><th>Low</th><th>Medium</th><th>High</th></tr><tr><td>Regular shares</td><td>7%</td><td>10%</td><td>15%</td></tr><tr><td>Risky shares</td><td>−10%</td><td>12%</td><td>25%</td></tr><tr><td>Property</td><td>−12%</td><td>18%</td><td>30%</td></tr></table> Past data: 300 days observed 150 days with medium market conditions 60 days with high market increases Task: State the optimum investment strategy for the investor.	Investment	Low	Medium	High	Regular shares	7%	10%	15%	Risky shares	−10%	12%	25%	Property	−12%	18%	30%	2020-II	3																																											
Investment	Low	Medium	High																																																													
Regular shares	7%	10%	15%																																																													
Risky shares	−10%	12%	25%																																																													
Property	−12%	18%	30%																																																													
	Q.41	Project Crashing Problem	The time & cost estimates of different activities are as given below. <table><tr><th rowspan="2">Activity (i-j)</th><th colspan="2">Time(Weeks)</th><th colspan="2">Cost(Rs.)</th></tr><tr><th>Normal</th><th>Crash</th><th>Normal</th><th>Crash</th></tr><tr><td>1-2</td><td>3</td><td>2</td><td>300</td><td>400</td></tr><tr><td>2-3</td><td>3</td><td>3</td><td>30</td><td>30</td></tr><tr><td>2-4</td><td>7</td><td>5</td><td>420</td><td>580</td></tr><tr><td>2-5</td><td>9</td><td>7</td><td>720</td><td>810</td></tr><tr><td>3-5</td><td>5</td><td>4</td><td>250</td><td>300</td></tr><tr><td>4-5</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>5-6</td><td>6</td><td>4</td><td>320</td><td>410</td></tr><tr><td>6-7</td><td>4</td><td>3</td><td>400</td><td>470</td></tr><tr><td>6-8</td><td>13</td><td>10</td><td>780</td><td>900</td></tr><tr><td>7-8</td><td>10</td><td>9</td><td>1000</td><td>1200</td></tr></table> Indirect cost is Rs. 50 per weeks. i) Draw the network & identify critical path. ii) Crash relevant activities systematically & find optimal project duration & cost	Activity (i-j)	Time(Weeks)		Cost(Rs.)		Normal	Crash	Normal	Crash	1-2	3	2	300	400	2-3	3	3	30	30	2-4	7	5	420	580	2-5	9	7	720	810	3-5	5	4	250	300	4-5	0	0	0	0	5-6	6	4	320	410	6-7	4	3	400	470	6-8	13	10	780	900	7-8	10	9	1000	1200	2020-I	6
Activity (i-j)	Time(Weeks)		Cost(Rs.)																																																													
	Normal	Crash	Normal	Crash																																																												
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4-5	0	0	0	0																																																												
5-6	6	4	320	410																																																												
6-7	4	3	400	470																																																												
6-8	13	10	780	900																																																												
7-8	10	9	1000	1200																																																												
	Q.42	CPM Network	Consider a CPM network with the following project data: <table><tr><th>Activity</th><th>Duration (days)</th><th>Predecessor(s)</th></tr><tr><td>A</td><td>4</td><td>None</td></tr><tr><td>B</td><td>3</td><td>None</td></tr><tr><td>C</td><td>6</td><td>A</td></tr><tr><td>D</td><td>2</td><td>B</td></tr><tr><td>E</td><td>5</td><td>C, D</td></tr><tr><td>F</td><td>3</td><td>D</td></tr><tr><td>G</td><td>7</td><td>A, B</td></tr><tr><td>H</td><td>5</td><td>E, F</td></tr></table> i) Draw a network diagram. Perform Forward Pass and Backward Pass calculations to determine the earliest start (ES), earliest finish (EF), latest start (LS) and latest	Activity	Duration (days)	Predecessor(s)	A	4	None	B	3	None	C	6	A	D	2	B	E	5	C, D	F	3	D	G	7	A, B	H	5	E, F	2024	5																																
Activity	Duration (days)	Predecessor(s)																																																														
A	4	None																																																														
B	3	None																																																														
C	6	A																																																														
D	2	B																																																														
E	5	C, D																																																														
F	3	D																																																														
G	7	A, B																																																														
H	5	E, F																																																														

			finish (LF) for each activity. ii) Calculate the total float and free float for each activity. Identify the activities that are on the critical path.																																																					
Q.43	Time Estimates	A project consists of 9 activities whose time estimates in weeks & other characteristic are given below: <table><tr><th>Activity</th><th>Preceding activity</th><th>most likely time (Week)</th><th>Pessimistic time (Week)</th><th>Optimistic time (Week)</th></tr><tr><td>A</td><td>---</td><td>4</td><td>6</td><td>2</td></tr><tr><td>B</td><td>---</td><td>6</td><td>6</td><td>6</td></tr><tr><td>C</td><td>---</td><td>12</td><td>24</td><td>6</td></tr><tr><td>D</td><td>A</td><td>5</td><td>8</td><td>2</td></tr><tr><td>E</td><td>A</td><td>14</td><td>23</td><td>11</td></tr><tr><td>F</td><td>B, D</td><td>10</td><td>12</td><td>8</td></tr><tr><td>G</td><td>B, D</td><td>6</td><td>9</td><td>3</td></tr><tr><td>H</td><td>C, F</td><td>15</td><td>27</td><td>9</td></tr><tr><td>I</td><td>E</td><td>10</td><td>16</td><td>4</td></tr></table>			Activity	Preceding activity	most likely time (Week)	Pessimistic time (Week)	Optimistic time (Week)	A	---	4	6	2	B	---	6	6	6	C	---	12	24	6	D	A	5	8	2	E	A	14	23	11	F	B, D	10	12	8	G	B, D	6	9	3	H	C, F	15	27	9	I	E	10	16	4	2024	5
Activity	Preceding activity	most likely time (Week)	Pessimistic time (Week)	Optimistic time (Week)																																																				
A	---	4	6	2																																																				
B	---	6	6	6																																																				
C	---	12	24	6																																																				
D	A	5	8	2																																																				
E	A	14	23	11																																																				
F	B, D	10	12	8																																																				
G	B, D	6	9	3																																																				
H	C, F	15	27	9																																																				
I	E	10	16	4																																																				
Q.44	CPM Network	A small project is composed of 7 activities whose time estimates are listed below. Activities are identified by their beginning (i) and ending (j) node numbers. <table><tr><th>i</th><th>j</th><th>t_o</th><th>t_i</th><th>t_p</th></tr><tr><td>1</td><td>2</td><td>1</td><td>4</td><td>7</td></tr><tr><td>1</td><td>3</td><td>1</td><td>2</td><td>8</td></tr><tr><td>1</td><td>4</td><td>2</td><td>2</td><td>8</td></tr><tr><td>2</td><td>5</td><td>1</td><td>5</td><td>14</td></tr><tr><td>4</td><td>6</td><td>2</td><td>5</td><td>8</td></tr><tr><td>5</td><td>6</td><td>3</td><td>6</td><td>15</td></tr></table> Tasks: a) Draw the network b) Calculate the expected variances for each c) Find the expected project completion time d) Calculate the probability that the project will be completed at least 3 weeks earlier than expected e) If the project due date is 18 weeks, what is the probability of not meeting the due date			i	j	t _o	t _i	t _p	1	2	1	4	7	1	3	1	2	8	1	4	2	2	8	2	5	1	5	14	4	6	2	5	8	5	6	3	6	15	2020-II	10															
i	j	t _o	t _i	t _p																																																				
1	2	1	4	7																																																				
1	3	1	2	8																																																				
1	4	2	2	8																																																				
2	5	1	5	14																																																				
4	6	2	5	8																																																				
5	6	3	6	15																																																				
Q.45	Floats of network Diagram	The utility data for a network are given below. Determine the total, free, independent, and interfering floats and identify the critical path. Activities: 0–1, 1–2, 1–3, 2–4, 2–5, 3–4, 4–7, 5–7, 6–7 Durations: 2, 8, 10, 6, 3, 3, 7, 5, 2, 8			2020-II	10																																																		
Q.46	Network Diagram, Critical Path, Probability	The following table shows the jobs of a network along with their time estimates: <table><tr><th>Jobs</th><th>Optimistic time (days)</th><th>Most likely time (days)</th><th>Pessimistic time (days)</th></tr><tr><td>1–2</td><td>1</td><td>7</td><td>13</td></tr><tr><td>1–6</td><td>2</td><td>5</td><td>14</td></tr><tr><td>2–3</td><td>2</td><td>14</td><td>26</td></tr><tr><td>2–4</td><td>4</td><td>5</td><td>8</td></tr><tr><td>3–5</td><td>7</td><td>10</td><td>19</td></tr><tr><td>4–5</td><td>5</td><td>5</td><td>17</td></tr><tr><td>6–7</td><td>5</td><td>8</td><td>29</td></tr><tr><td>5–8</td><td>3</td><td>3</td><td>9</td></tr><tr><td>7–8</td><td>8</td><td>17</td><td>32</td></tr></table> Tasks: i) Draw the network diagram ii) Find the critical path iii) Project completion time and project variance iv) Probability that project will be completed in 40 days			Jobs	Optimistic time (days)	Most likely time (days)	Pessimistic time (days)	1–2	1	7	13	1–6	2	5	14	2–3	2	14	26	2–4	4	5	8	3–5	7	10	19	4–5	5	5	17	6–7	5	8	29	5–8	3	3	9	7–8	8	17	32	2019	10										
Jobs	Optimistic time (days)	Most likely time (days)	Pessimistic time (days)																																																					
1–2	1	7	13																																																					
1–6	2	5	14																																																					
2–3	2	14	26																																																					
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3–5	7	10	19																																																					
4–5	5	5	17																																																					
6–7	5	8	29																																																					
5–8	3	3	9																																																					
7–8	8	17	32																																																					

5-Decision Theory & Game Theory	Q.47	Decision Theory -Decision Making Under Risk	A major consumer goods manufacturer wishes to decide new product bring out in the market which of the profit table is as follows (Profit per in units of Rs. 10000). <table><tr><th rowspan="2">Strategies</th><th colspan="3">States</th></tr><tr><th>S₁(0.4)</th><th>S₂(0.5)</th><th>S₃(0.1)</th></tr><tr><td>A1</td><td>140</td><td>100</td><td>80</td></tr><tr><td>A2</td><td>160</td><td>130</td><td>120</td></tr><tr><td>A3</td><td>200</td><td>160</td><td>140</td></tr></table> Determine:- i) Expected monetary value (EMV). ii) Expected monetary value with-perfect information (EMVPI). iii) Value of perfect information.(VPI)	Strategies	States			S ₁ (0.4)	S ₂ (0.5)	S ₃ (0.1)	A1	140	100	80	A2	160	130	120	A3	200	160	140	2020-I	6					
	Strategies	States																											
		S ₁ (0.4)	S ₂ (0.5)	S ₃ (0.1)																									
	A1	140	100	80																									
	A2	160	130	120																									
A3	200	160	140																										
Q.48	Game Theory-Without saddle point	Solve the following game: <table><tr><th>Player A \ Player B</th><th>B1</th><th>B2</th></tr><tr><td>A1</td><td>3</td><td>5</td></tr><tr><td>A2</td><td>4</td><td>1</td></tr></table>	Player A \ Player B	B1	B2	A1	3	5	A2	4	1	2020-I	4																
Player A \ Player B	B1	B2																											
A1	3	5																											
A2	4	1																											
Q.49	Decision Theory-Decision Making Under Uncertainty	In a toy manufacturing company suppose product acceptance probability are not known but following data known. Anticipated 1st year profit (Rs. ‘000’) <table><tr><th>Product \ Acceptance</th><th>Full</th><th>Partial</th><th>Minimal</th></tr><tr><td>Good</td><td>8</td><td>70</td><td>50</td></tr><tr><td>Fair</td><td>50</td><td>45</td><td>40</td></tr><tr><td>Poor</td><td>-25</td><td>-10</td><td>0</td></tr></table> Determine optimal decision under: i) Maximax ii) Maximin iii) Minimax regret criteria iv)	Product \ Acceptance	Full	Partial	Minimal	Good	8	70	50	Fair	50	45	40	Poor	-25	-10	0	2020-I	4									
Product \ Acceptance	Full	Partial	Minimal																										
Good	8	70	50																										
Fair	50	45	40																										
Poor	-25	-10	0																										
Q.50	Decision Making Under Risk (EMV)	The following table gives profit matrix for different strategy and actions. Find the optimal strategy by using EMV Criterion <table><tr><th>States of Nature</th><th>Probability</th><th>A1</th><th>A2</th><th>A3</th></tr><tr><td>E1</td><td>0.15</td><td>40</td><td>52</td><td>45</td></tr><tr><td>E2</td><td>0.25</td><td>70</td><td>28</td><td>40</td></tr><tr><td>E3</td><td>0.45</td><td>30</td><td>70</td><td>-50</td></tr><tr><td>E4</td><td>0.15</td><td>30</td><td>-50</td><td>-70</td></tr></table>	States of Nature	Probability	A1	A2	A3	E1	0.15	40	52	45	E2	0.25	70	28	40	E3	0.45	30	70	-50	E4	0.15	30	-50	-70	2024	5
States of Nature	Probability	A1	A2	A3																									
E1	0.15	40	52	45																									
E2	0.25	70	28	40																									
E3	0.45	30	70	-50																									
E4	0.15	30	-50	-70																									
Q.51	Decision Making Under Uncertainty (Hurwitz Criterion, Savage Criterion)	A given Cost matrix, suggest optimum strategy using– <table><tr><th>Strategies</th><th>N1</th><th>N2</th><th>N3</th><th>N4</th></tr><tr><td>S1</td><td>1000</td><td>1500</td><td>750</td><td>0</td></tr><tr><td>S2</td><td>250</td><td>2000</td><td>3750</td><td>3000</td></tr><tr><td>S3</td><td>-500</td><td>1250</td><td>3000</td><td>4750</td></tr><tr><td>S4</td><td>-1250</td><td>500</td><td>2250</td><td>4000</td></tr></table> i) Hurwitz Criterion (Alpha = 0.7) ii) Savage Criterion	Strategies	N1	N2	N3	N4	S1	1000	1500	750	0	S2	250	2000	3750	3000	S3	-500	1250	3000	4750	S4	-1250	500	2250	4000	2024	5
Strategies	N1	N2	N3	N4																									
S1	1000	1500	750	0																									
S2	250	2000	3750	3000																									
S3	-500	1250	3000	4750																									
S4	-1250	500	2250	4000																									

	Q.52	Dominance Method(Without Saddle Point)	Define a zero-sum game and explain the concept of dominance in game theory. Solve the following 2×2 zero-sum game using dominance: <table><tr><th>Player</th><th>Strategy 1</th><th>Strategy 2</th></tr><tr><td>Player 1</td><td>6</td><td>4</td></tr><tr><td>Player 2</td><td>5</td><td>3</td></tr></table> Identify the dominated strategy (if any) for both players. Simplify the game using the dominance rule and find the optimal strategies for both players.	Player	Strategy 1	Strategy 2	Player 1	6	4	Player 2	5	3	2024	5																					
Player	Strategy 1	Strategy 2																																	
Player 1	6	4																																	
Player 2	5	3																																	
	Q.53	Dominance Method(Without Saddle Point)	Solve the following to find the value of the game <table><tr><th>Player A</th><th>B1</th><th>B2</th><th>B3</th><th>B4</th><th>B5</th></tr><tr><td>A1</td><td>2</td><td>4</td><td>3</td><td>8</td><td>7</td></tr><tr><td>A2</td><td>5</td><td>6</td><td>3</td><td>7</td><td>8</td></tr><tr><td>A3</td><td>4</td><td>5</td><td>8</td><td>8</td><td>7</td></tr><tr><td>A4</td><td>4</td><td>2</td><td>8</td><td>4</td><td>3</td></tr></table> Would you like me to solve one of these problems step by step (e.g., the EMV criterion or the dominance game), or just keep it as a clean extraction?	Player A	B1	B2	B3	B4	B5	A1	2	4	3	8	7	A2	5	6	3	7	8	A3	4	5	8	8	7	A4	4	2	8	4	3	2024	5
Player A	B1	B2	B3	B4	B5																														
A1	2	4	3	8	7																														
A2	5	6	3	7	8																														
A3	4	5	8	8	7																														
A4	4	2	8	4	3																														
	Q.54	Mixed Strategy	Solve the given game by method of matrices: <table><tr><td></td><td>B I</td><td>B II</td><td>B III</td></tr><tr><td>A I</td><td>1</td><td>−1</td><td>−1</td></tr><tr><td>A II</td><td>−1</td><td>−1</td><td>3</td></tr><tr><td>A III</td><td>−1</td><td>2</td><td>−1</td></tr></table>		B I	B II	B III	A I	1	−1	−1	A II	−1	−1	3	A III	−1	2	−1	2020-II	7														
	B I	B II	B III																																
A I	1	−1	−1																																
A II	−1	−1	3																																
A III	−1	2	−1																																
	Q.55	Dominance Method, Mixed Strategy (Without Saddle Point)	A company is currently involved in negotiations with its union on the upcoming wage contract. Positive signs in the table represent wage increase, while negative signs represent wage reduction. Task: - What are the optimal strategies for the company as well as the union? - What is the game value? Conditional costs to the company (Rs. in lakhs): <table><tr><th>Company Strategies</th><th>U₁</th><th>U₂</th><th>U₃</th><th>U₄</th></tr><tr><td>C₁</td><td>0.25</td><td>0.27</td><td>0.35</td><td>-0.02</td></tr><tr><td>C₂</td><td>0.20</td><td>0.06</td><td>0.08</td><td>0.08</td></tr><tr><td>C₃</td><td>0.14</td><td>0.12</td><td>0.05</td><td>0.03</td></tr><tr><td>C₄</td><td>0.30</td><td>0.14</td><td>0.19</td><td>0.00</td></tr></table>	Company Strategies	U ₁	U ₂	U ₃	U ₄	C ₁	0.25	0.27	0.35	-0.02	C ₂	0.20	0.06	0.08	0.08	C ₃	0.14	0.12	0.05	0.03	C ₄	0.30	0.14	0.19	0.00	2020-II	7					
Company Strategies	U ₁	U ₂	U ₃	U ₄																															
C ₁	0.25	0.27	0.35	-0.02																															
C ₂	0.20	0.06	0.08	0.08																															
C ₃	0.14	0.12	0.05	0.03																															
C ₄	0.30	0.14	0.19	0.00																															
	Q.56	Decision Making under Uncertainty (Minimax Criterion)	A manufacturer produces a product whose principal ingredient is chemical X. - Current annual cost of X: Rs. 1,000 - Possible price increase: up to 4 times due to worldwide shortage Alternative option: - Use chemical Y with chemical Z together to replace X - Annual cost of Y + Z: Rs. 3,000 - Prices of Y and Z are stable Task: What action should the manufacturer take using minimax criteria for decision-making?	2020-II	3																														



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	Q.57	Dominance Method	Solve the following game. Find the optimal strategy of A and B, and the value of the game. Payoff Matrix: <table border="1"> <tr> <th>Player A</th><th>B₁</th><th>B₂</th><th>B₃</th><th>B₄</th><th>B₅</th></tr> <tr> <td>A₁</td><td>9</td><td>3</td><td>1</td><td>8</td><td>0</td></tr> <tr> <td>A₂</td><td>6</td><td>5</td><td>4</td><td>6</td><td>7</td></tr> <tr> <td>A₃</td><td>4</td><td>5</td><td>4</td><td>3</td><td>8</td></tr> <tr> <td>A₄</td><td>5</td><td>6</td><td>2</td><td>2</td><td>1</td></tr> </table>	Player A	B ₁	B ₂	B ₃	B ₄	B ₅	A ₁	9	3	1	8	0	A ₂	6	5	4	6	7	A ₃	4	5	4	3	8	A ₄	5	6	2	2	1	2019	10
Player A	B ₁	B ₂	B ₃	B ₄	B ₅																														
A ₁	9	3	1	8	0																														
A ₂	6	5	4	6	7																														
A ₃	4	5	4	3	8																														
A ₄	5	6	2	2	1																														

Course faculty